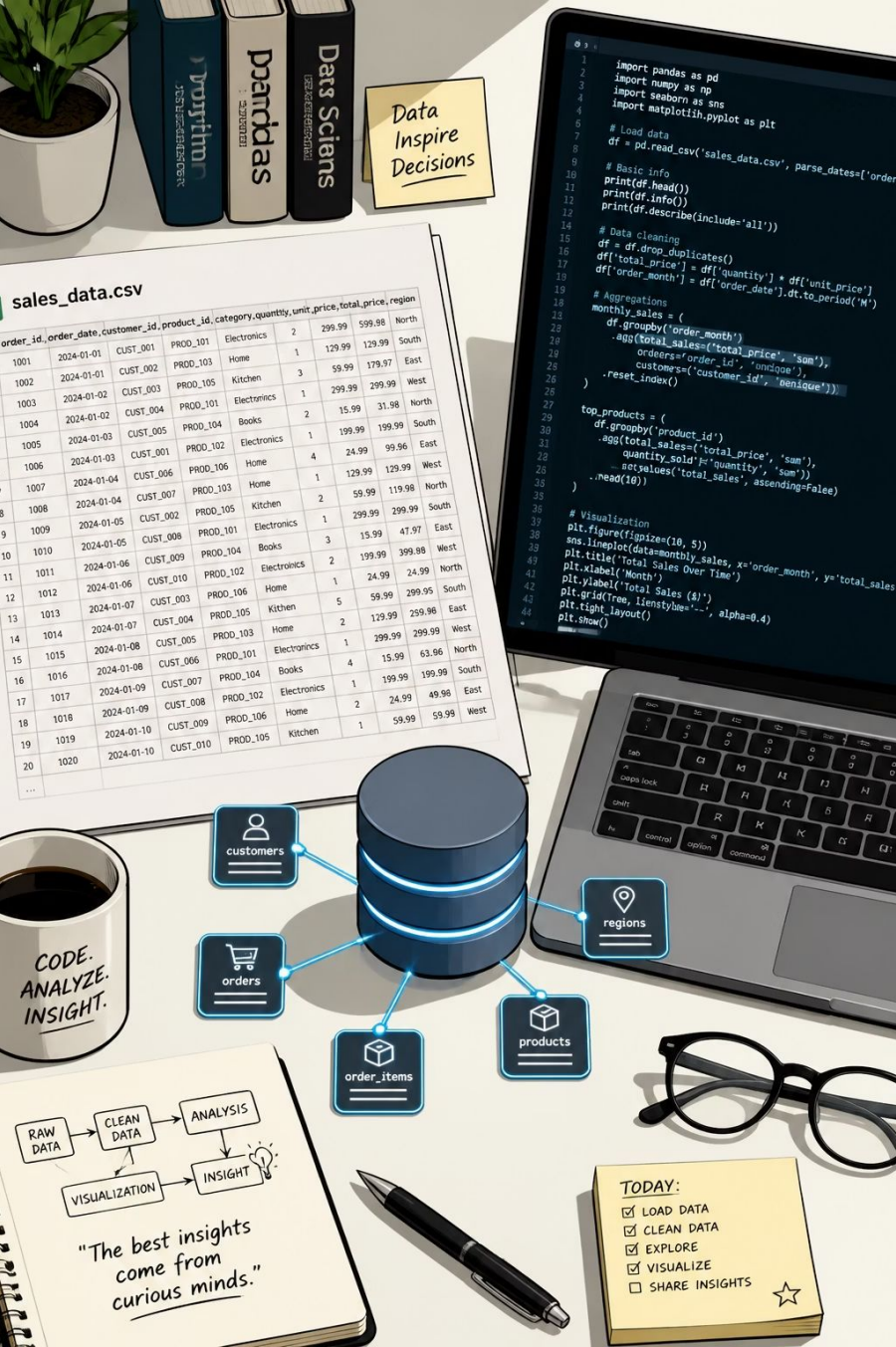




Customer Shopping Behavior Analysis

End-to-end pipeline: Pandas → PostgreSQL → SQL insights → Power BI



Project Overview

Dataset

customer_shopping_behavior.cs
v — 3,900 records · 18 columns

Scope

Cleaning, DB load, 10 SQL
questions, Power BI dashboard

Goal

Understand behavior, revenue, product & segment patterns

Data Cleaning & Prep (Pandas)

Inspect

`.info(), .describe()`

Impute

Filled missing `review_rating` with category median

Normalize

Standardized column names to lowercase_with_underscores

Feature

`age_group` via quartile binning; `purchase_frequency_days` numeric

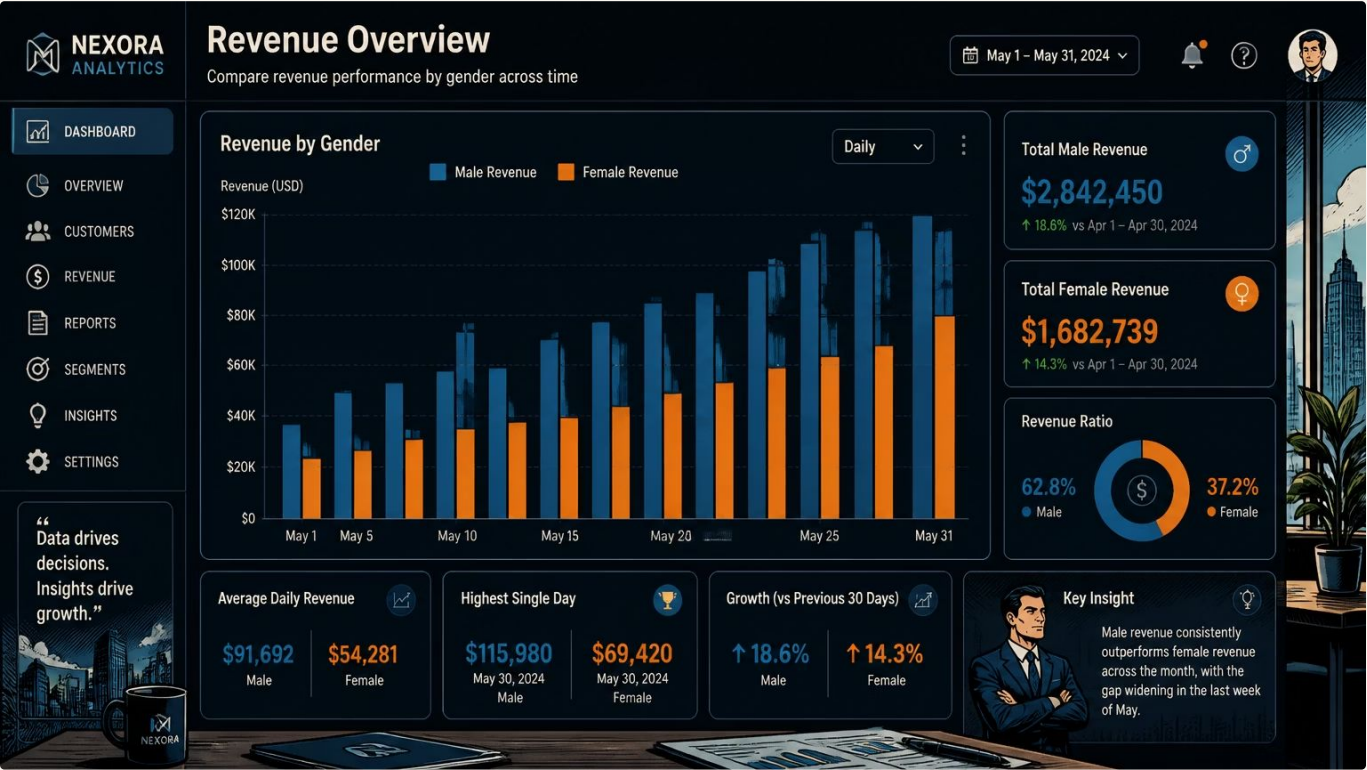
Prune

Dropped redundant `promo_code_used` (same as `discount_applied`)

Load

to PostgreSQL using SQLAlchemy `to_sql()`

Key Insight — Revenue by Gender



Finding

Male total revenue: \$157,890

Female total revenue: \$75,191

☐ Males generated significantly more revenue

Discounts & Spending



839 Discount High-Spenders

Used discount but spent > average
(\$59.76)

Top Discounted Products

Hat 50.0% · Sneakers 49.7% · Coat
49.1% · Sweater 48.2% · Pants 47.4%

Product Ratings & Top Items



Gloves
Avg rating 3.86



Sandals
Avg rating 3.84



Boots
Avg rating 3.82

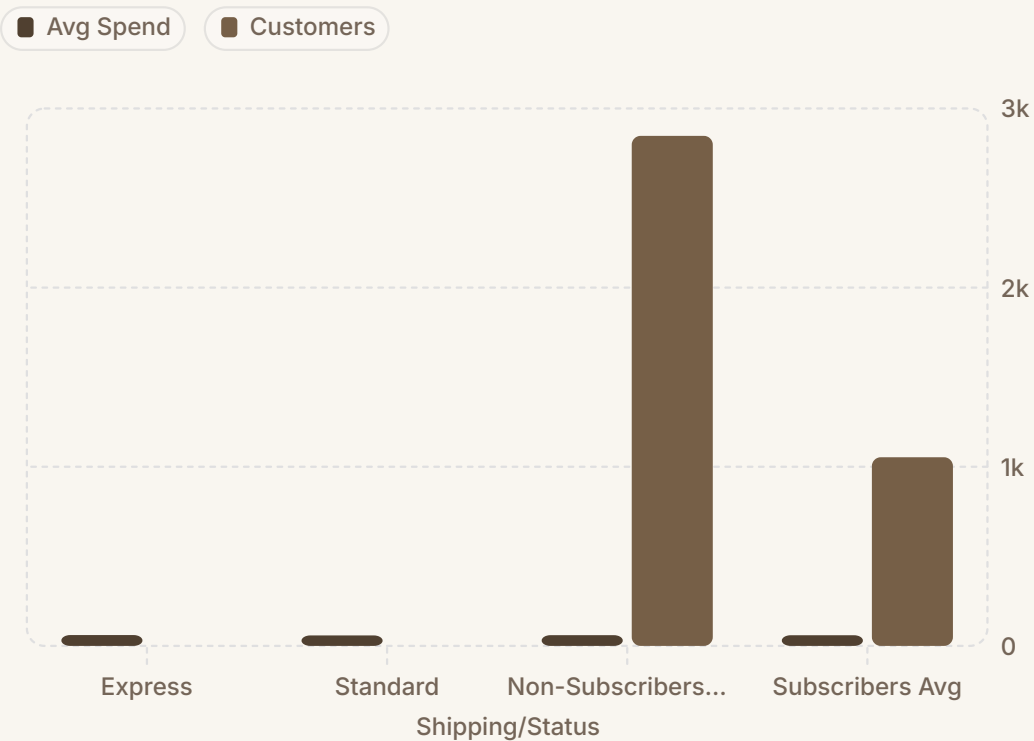


Hat
Avg rating 3.80



Skirt
Avg rating 3.79

Shipping & Subscription Effects



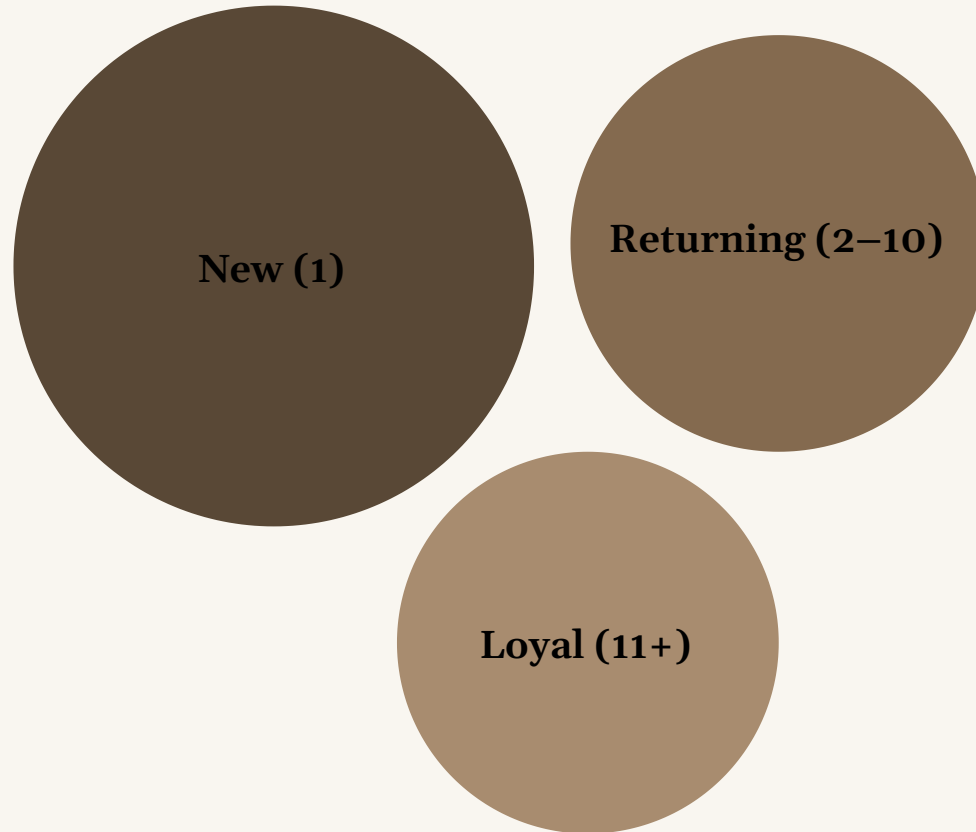
Express slightly higher avg spend than Standard.

Non-subscribers: 2,847 customers · Avg \$59.87 · Total \$170,436

Subscribers: 1,053 customers · Avg \$59.49 · Total \$62,645

📌 Non-subscribers contribute more total revenue due to volume

Customer Segmentation & Loyalty



Loyal

3,116 customers ($\approx 80\%$)

Returning

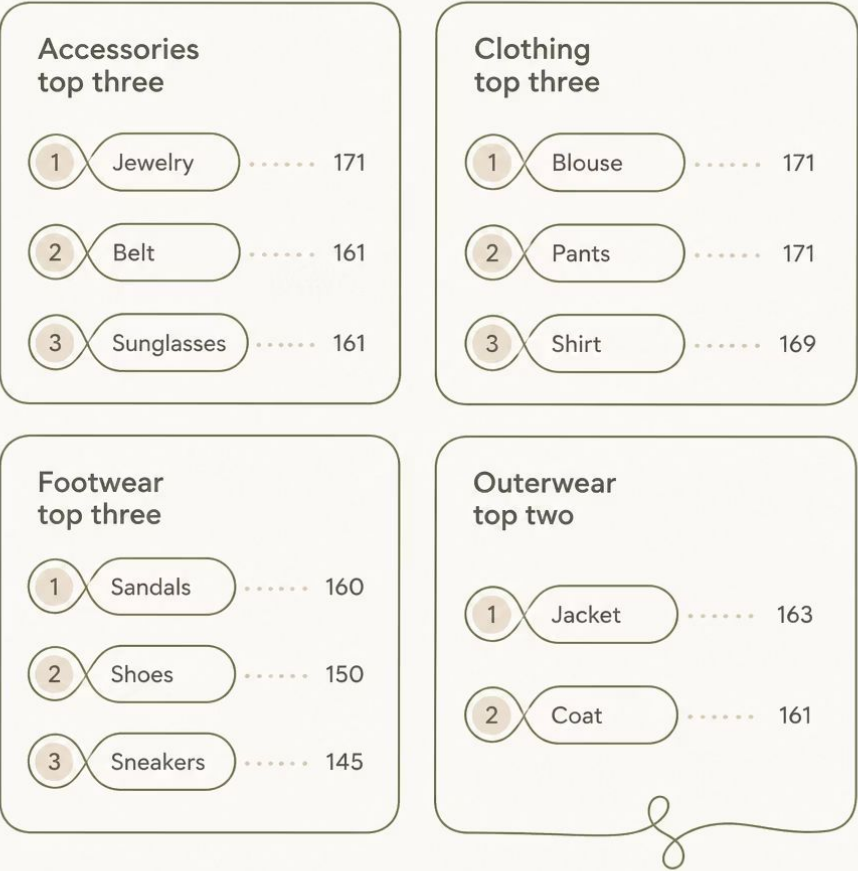
701 customers

New

83 customers

Most repeat buyers are non-subscribers:
2,518 non-subscriber repeat vs 958 subscribers

Product Performance by Category



Category winners highlight product focus areas and inventory priorities.

Accessories

Jewelry (171) · Belt (161) ·
Sunglasses (161)

Clothing

Blouse (171) · Pants (171) ·
Shirt (169)

Footwear

Sandals (160) · Shoes (150) · Sneakers (145)

Final Dashboard & Tech Stack



Data Cleaning

Python (Pandas) · Jupyter



Database

PostgreSQL (customer_behavior)



Analysis

SQL — 10 business questions answered



Visualization

Power BI interactive dashboard

- ❑ Dashboard enables stakeholders to explore revenue, segments, product and discount behavior